THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE UNIT TEST SET B

Chapters 9–12: **Light – Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The refractive index of water is 1.33. The speed of light in water is about: 1

(a) 3×10^8 m/s **(b)** 1.33×10^8 m/s **(c)** 4×10^8 m/s **(d)** 2.25×10^8 m/s

2. The radius of curvature of a concave mirror is 20 cm. Its focal length is: 1

(a) +10 cm **(b)** −20 cm **(c)** −40 cm **(d)** −10 cm

3. The mirror used by a dentist is: 1

(a) convex **(b)** cylindrical **(c)** plane **(d)** concave

4. The ability of the eye lens to adjust its focal length is called: 1

(a) persistence of vision **(b)** power of lens
(c) accommodation **(d)** dispersion

5. The defect in which a person cannot see nearby objects clearly is: 1

(a) presbyopia only **(b)** myopia **(c)** cataract **(d)** hypermetropia

6. The change in focal length of the eye lens is caused by: 1

(a) cornea **(b)** pupil **(c)** retina **(d)** ciliary muscles

7. The SI unit of electric current is: 1

(a) ampere **(b)** volt **(c)** ohm **(d)** coulomb

8. A student states: “The resistance of a wire depends on the material.” The correct response is: 1

(a) Correct **(b)** Only on length
(c) Incorrect **(d)** Cannot be determined

9. Work done in moving 2 C of charge through 12 V is: 1

(a) 24 J **(b)** 0.17 J **(c)** 14 J **(d)** 6 J

10. Magnetic field lines emerge from the: 1

(a) south pole and enter at the north pole **(b)** middle of the magnet
(c) north pole and enter at the south pole **(d)** nowhere

11. The magnetic field inside a long current-carrying solenoid is: 1

(a) uniform **(b)** maximum at the ends
(c) zero **(d)** circular

12. The colour of the insulation of the earth wire is usually: 1

(a) blue **(b)** black **(c)** green **(d)** red

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 10–15 words each.

13. Why does a pencil appear bent when partly immersed in water? 2

14. What is the Tyndall effect? 2

15. Calculate the current through a 40 Ω resistor connected to a 12 V battery. 2

16. A student says that household appliances are connected in series. Justify or refute. 2

17. What is a magnetic field? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. An object is placed 10 cm in front of a convex mirror of focal length 15 cm. Find the position and magnification of the image. 3

19. What is hypermetropia? Give two causes and its correction. 3

20. An electric bulb is rated 100 W, 220 V. Find 3
(i) its resistance
(ii) the current drawn
(iii) energy consumed in 10 h.

21. Compare the field due to a straight conductor and a solenoid. 3

22. What is the relation between focal length and radius of curvature? Find R for a mirror with $f = -12$ cm. 3

OR

Define 3
(i) refractive index
(ii) optical centre
(iii) principal axis.

23. Explain advanced sunrise and delayed sunset. 3

OR

Draw (describe) the path of a ray of light through a glass prism. Label the angle of deviation. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. **Case study:** A student traced the path of a ray through a rectangular glass slab and found the emergent ray parallel to the incident ray but shifted. 5
(i) Why is the emergent ray parallel?
(ii) What is the shift called?
(iii) What happens to speed of light in glass?
(iv) If $n = 1.5$, find the speed of light in glass.

OR

(i) What is meant by refraction? 5
(ii) Light enters from air into water ($n = 4/3$). Find the speed of light in water.
(iii) What is the refractive index of water with respect to glass ($n_g = 3/2$)?

25. (i) Derive the expression for heat produced: $H = I^2Rt$. 5
(ii) An electric kettle of resistance 50 Ω draws 4 A. Find the heat produced in 1 minute.

OR

(i) Explain the dependence of the resistance of a conductor on length and area. 5
(ii) A copper wire has diameter 0.5 mm and resistivity 1.6×10^{-8} Ω m. Find the length for 10 Ω resistance.

END OF QUESTION PAPER — Total: 50 Marks

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